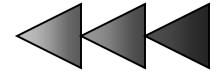


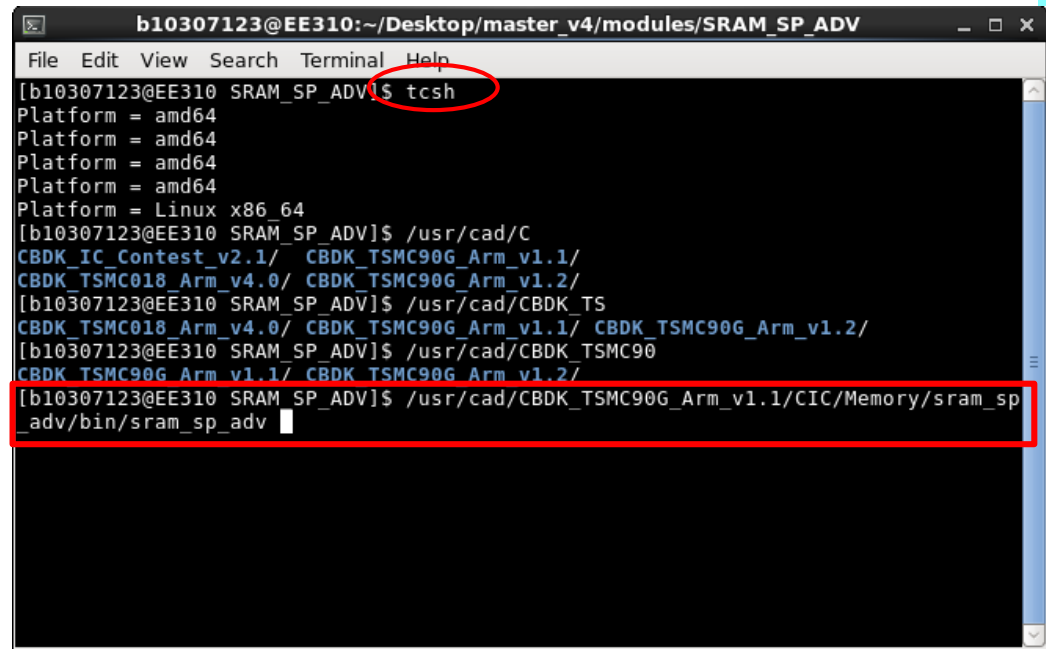
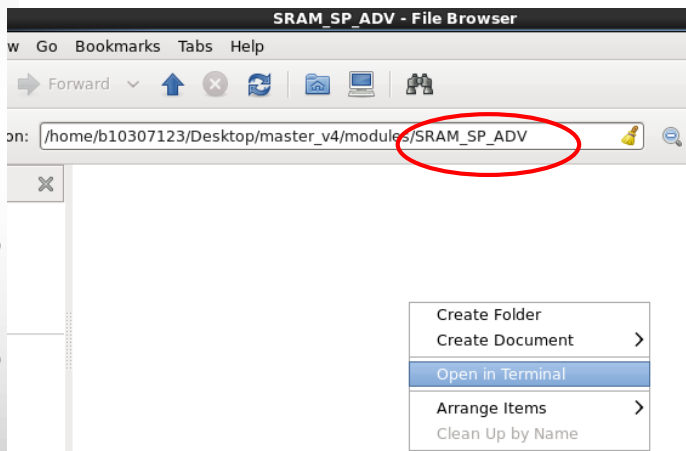
Memory Generator

目前只能在舊伺服器產Memory!!!

Memory Generation



- ❑ If one is desired to generate a SRAM macro, open a terminal at ones folder and enter (creating a new memory folder is optional but recommended):



ARM

File Utilities Help

Artisan[®]
ARM Physical IP

Advantage Single-Port SRA
TSMC 90nm CLN90G F

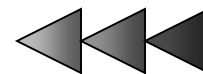
GENERIC PARAMETERS

Instance Name	SRAM_SP_ADV
Number of Words	16384 地址數
Number of Bits	8 一個地址位元數
Frequency <MHz>	250 工作頻率
Ring Width <um>	2.0 供電線寬度
Multiplexer Width	☐ 8 ☐ 16 ☒ 32
Word-Write Mask	☐ on ☒ off
Top Metal Layer	☒ m5-9
Power Type	☒ rings ☐ ArtiGrid
Horizontal Ring Layer	☐ m2 ☒ m3 ☐ m4
Vertical Ring Layer	☐ m2 ☐ m3 ☒ m4
Redundancy	☐ on ☒ off
BIST MUXes	☐ on ☒ off
Soft Error Repair (SER)	☒ none ☐ 1bd1bc ☐ 2bd1bc
Extra Margin Adjustment	☒ on

Default Update

Table 3-17. 90nm Single-Port SRAM (sram_sp_adv) Parameters

Input Parameters		
Parameter	Ranges	
number of words	mux = 8	256 to 4096, increment = mux • 4
	mux = 16	512 to 8192, increment = mux • 4
	mux = 32	1024 to 16384, increment = mux • 4
number of bits	mux = 8	2 to 128, increment = 1
	mux = 16	2 to 64, increment = 1
	mux = 32	2 to 32, increment = 1
frequency	1 to 1/t _{cyc} • 1000 , increment = 1	
word partition size ¹	1 to min (36, bits-1) increment = 1	
multiplexer width (μm)	8, 16, 32; default 16	
word-write mask	on, off; default off	
top chip metal layer support	m4 to top metal layer supported by design process	
top generator metal layer	m5 - 9	
power type	Rings, ArtiGrid (OTC); default rings	
horizontal ring layer	m2, m3, or m4; Ring power type must be selected; default m3	
vertical ring layer	m2, m3, or m4; Ring power type must be selected; default m4	
ring width	2 to 10,000	
redundancy	on, off; default off	
redundant columns	2; redundancy must be selected	
redundant rows	2; redundancy must be selected	
BIST MUXes	on, off; default off	
soft error repair (SER) ²	none, 1bd1bd, 2bd1bc; default none	
Block Parameters		
Parameter	Ranges	
total memory bits	512 to 524,288, total bits = words • bits	
rows in memory matrix	32 to 512, increment = 4, rows = words / mux	
columns in memory matrix	16 to 1024, increment = mux, columns = bits • mux	
address lines	mux = 8	8 to 12
	mux = 16	9 to 13
	mux = 32	10 to 14
output drive strength, Q[]	See "Running the Generator from the Graphical User Interface (GUI)" on page 2-3 for generator GUI or command line help instructions	



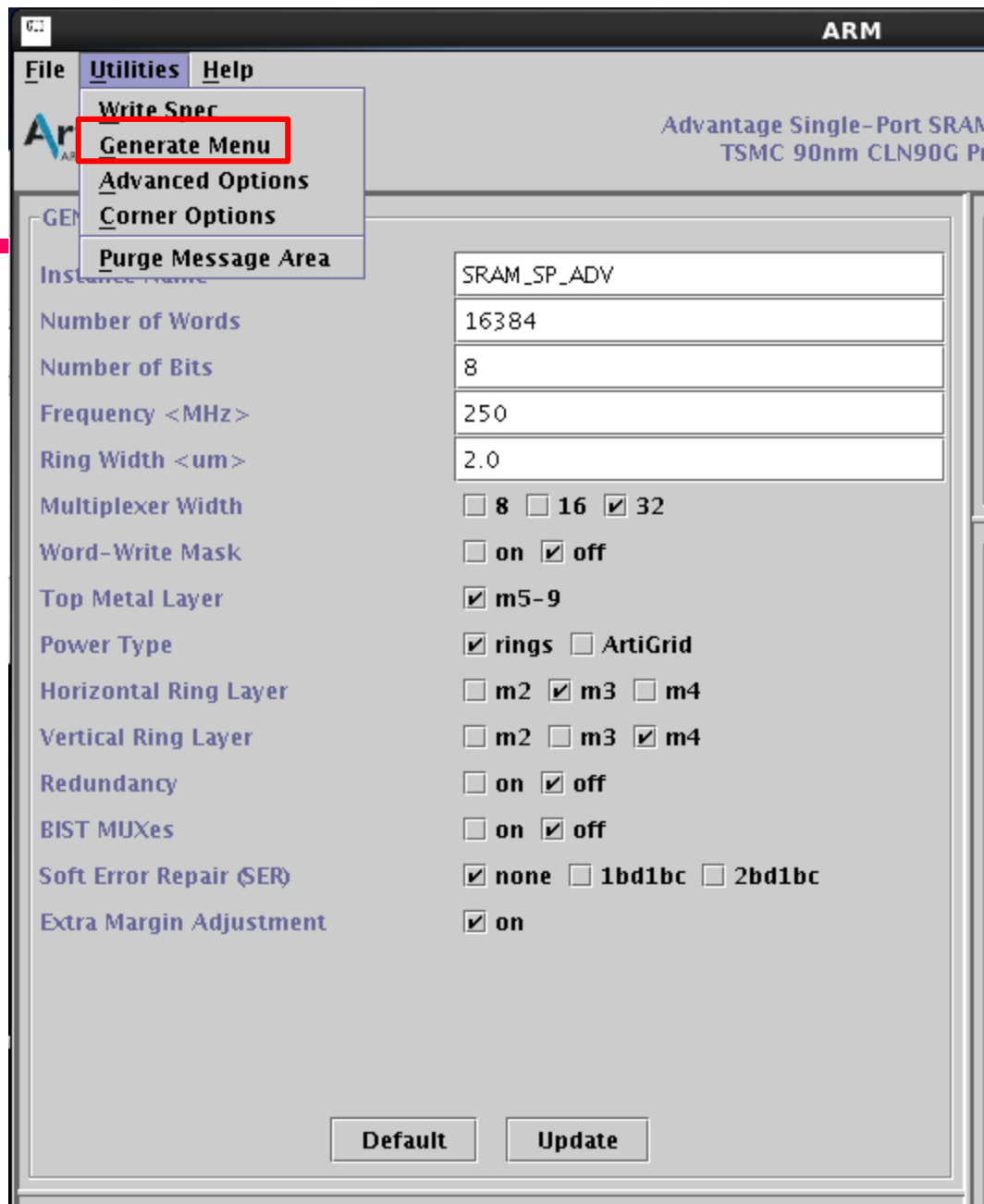
根據自己所需memory
規格選擇對應 mux

Row
Addr.

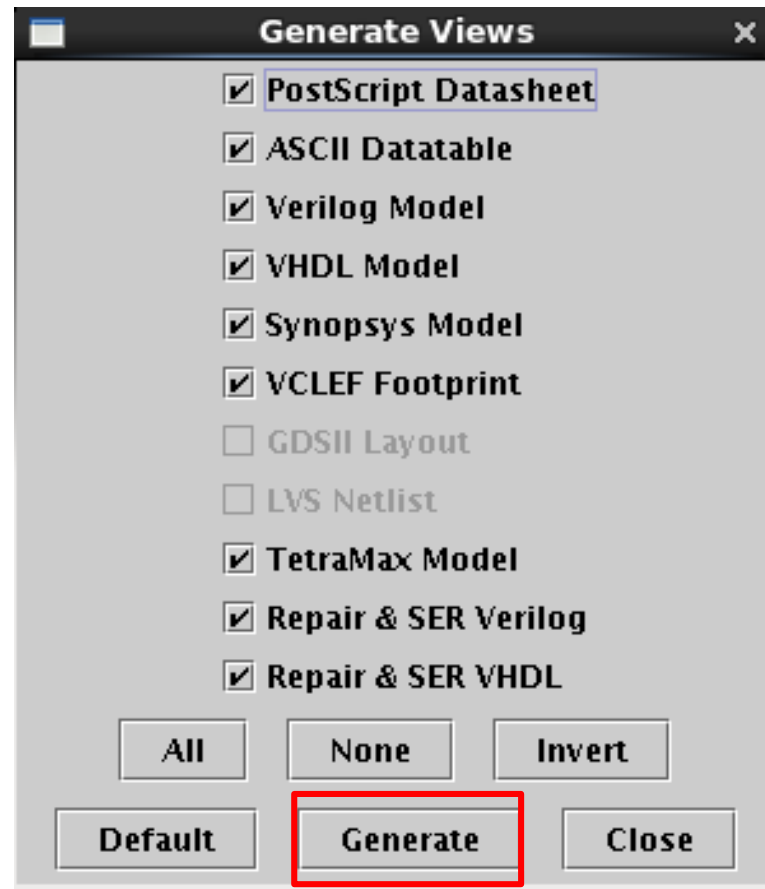
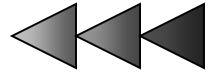
Col.
Addr.

Row Addr. Length =
 $\log_2 (\text{words}/\text{mux})$

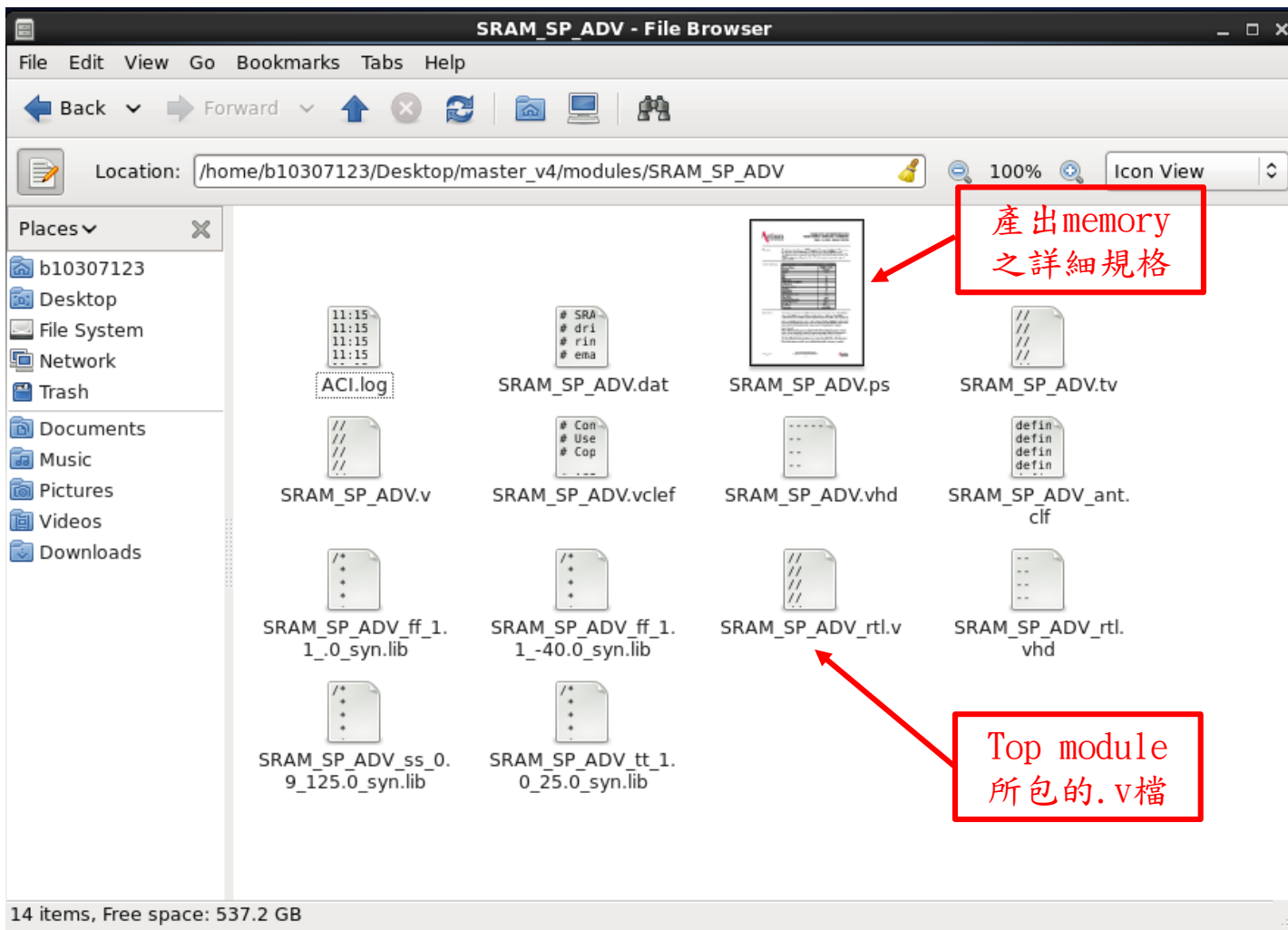
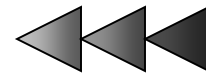
Col. Addr. Length =
 $\log_2 (\text{mux})$



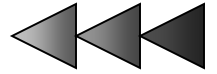
Memory Generation



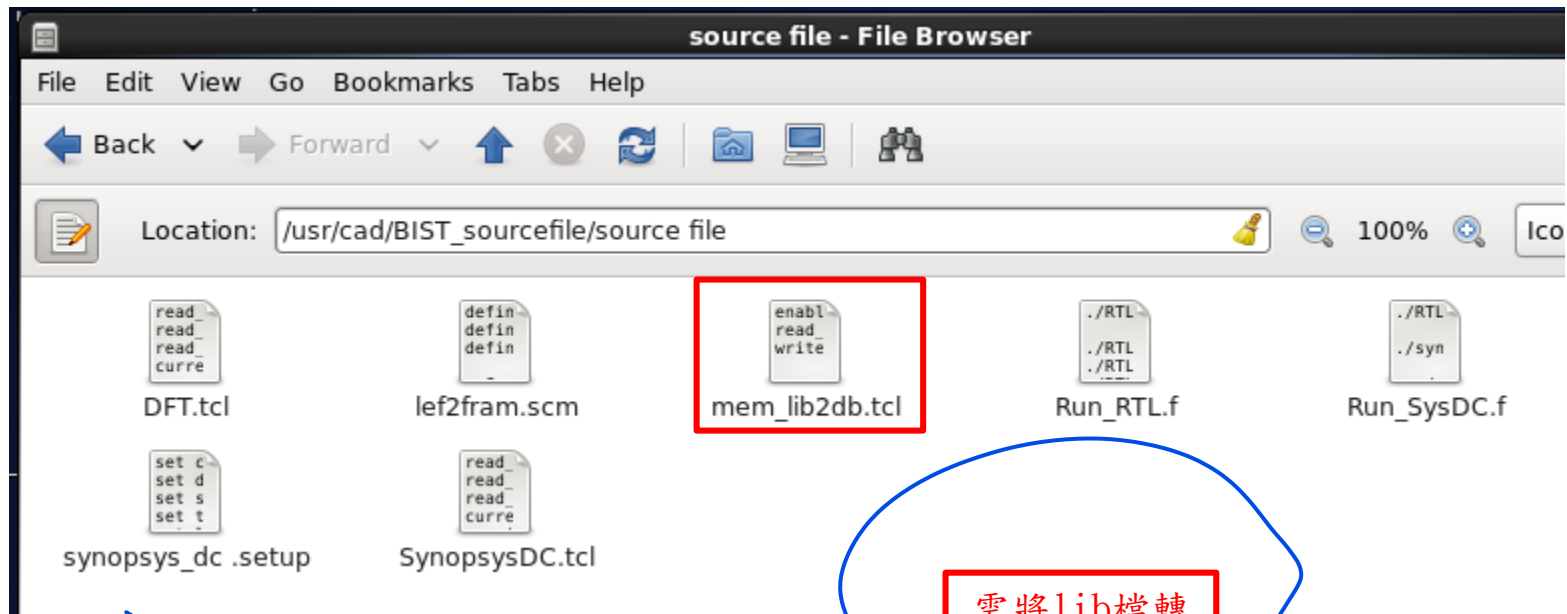
Memory Generation



Generate memory db file

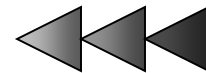


- ❑ Go to `/usr/cad/BIST_sourcefile/source file` and copy `mem_lib2db.tcl` file into ur memory directory



問一下

Generate memory db file



The screenshot shows an Emacs editor window titled "mem_lib2db.tcl - emacs@EE310" with the following code:

```
enable write lib mode
read_lib /home/aaazx610/Desktop/layout/SRAM_SP_ADV/SRAM_SP_ADV_ss_0.9_125.0_syn.lib
write_lib USERLIB -output SRAM_SP_ADV_ss_0.9_125.0_syn.db
```

Below the code, the filename "SRAM_SP_ADV_ss_0.9_125.0_syn.db" is highlighted in blue. A red box on the left contains the text:

更改mem_lib2db.tcl內的
lib絕對地址
以及設定產出之db檔檔名

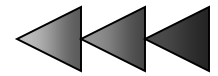
Below this, the text reads:

Edit tcl file to Make sure that the file
(**XXX_ss_0.9_125.0_syn.lib**) exists
編輯所屬路徑與檔案名稱

On the right, a file browser window shows a directory containing various files. A red box highlights the file "mem_lib2db.tcl" with the label "打開" (Open) in red. A blue box highlights the file "SRAM_SP_ADV_ss_0.9_125.0_syn.lib". A blue arrow points from the highlighted file in the browser to the "read_lib" line in the Emacs code. A red arrow points from the "mem_lib2db.tcl" file in the browser to the "write_lib" line in the Emacs code.

At the bottom of the browser window, the status bar shows: "SRAM_SP_ADV_ss_0.9_125.0_syn.lib" selected (69.6 KB).

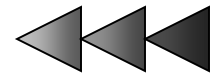
Generate memory db file



❑ Go back to ur terminal:

```
b10307123@EE310:~/Desktop/master_v4/modules/SRAM_SP_ADV
File Edit View Search Terminal Help
[b10307123@EE310 SRAM_SP_ADV]$ tcsh
Platform = amd64
Platform = amd64
Platform = amd64
Platform = amd64
Platform = Linux x86_64
[b10307123@EE310 SRAM_SP_ADV]$ /usr/cad/C
CBDK_IC_Constest_v2.1/ CBDK_TSMC90G_Arm_v1.1/
CBDK_TSMC018_Arm_v4.0/ CBDK_TSMC90G_Arm_v1.2/
[b10307123@EE310 SRAM_SP_ADV]$ /usr/cad/CBDK_TS
CBDK_TSMC018_Arm_v4.0/ CBDK_TSMC90G_Arm_v1.1/ CBDK_TSMC90G_Arm_v1.2/
[b10307123@EE310 SRAM_SP_ADV]$ /usr/cad/CBDK_TSMC90
CBDK_TSMC90G_Arm_v1.1/ CBDK_TSMC90G_Arm_v1.2/
[b10307123@EE310 SRAM_SP_ADV]$ /usr/cad/CBDK_TSMC90G_Arm_v1.1/CIC/Memory/sram_sp
adv/bin/sram_sp_adv
Loading, please wait...
[b10307123@EE310 SRAM_SP_ADV]$ dv -no_gui
```

Generate memory db file



❑ Go back to ur termial:

```
b10307123@EE310:~/Desktop/master_v4/modules/SRAM_SP_ADV
File Edit View Search Terminal Help

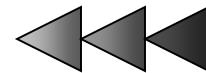
Design Compiler Graphical
DC Ultra (TM)
DFTMAX (TM)
Power Compiler (TM)
DesignWare (R)
DC Expert (TM)
Design Vision (TM)
HDL Compiler (TM)
VHDL Compiler (TM)
DFT Compiler
Library Compiler (TM)
Design Compiler(R)

Version J-2014.09-SP1 for RHEL64 -- Oct 15, 2014
Copyright (c) 1988-2014 Synopsys, Inc.

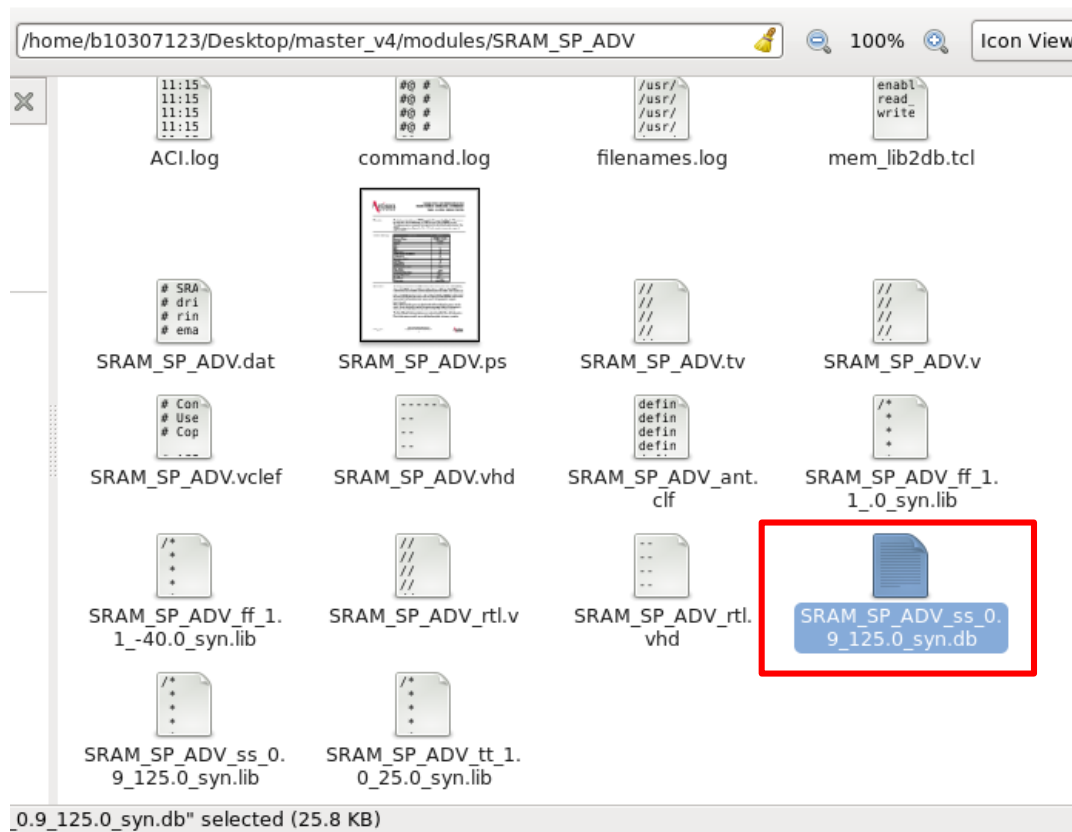
This software and the associated documentation are confidential and
proprietary to Synopsys, Inc. Your use or disclosure of this software
is subject to the terms and conditions of a written license agreement
between you, or your company, and Synopsys, Inc.

Initializing...
Initializing gui preferences from file /home/b10307123/.synopsys_dv_prefs.tcl
design_vision> source mem_lib2db.tcl
```

Generate memory db file

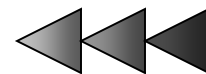


Generated db file



檢查是否順利產出db檔

Exit Design Vision

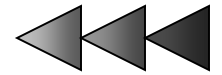


❑ Ctrl + Z or exit

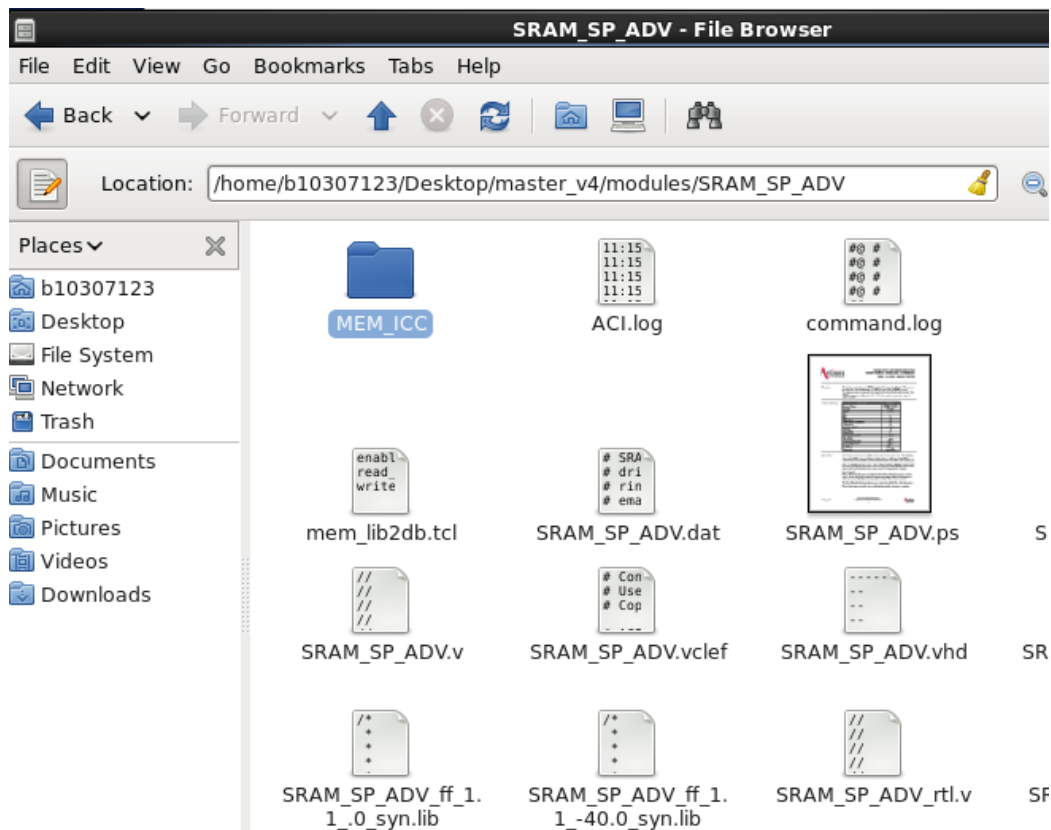
關掉design compiler

```
b10307123@EE310:~/Desktop/master_v4/modules/SRAM_SP_ADV
File Edit View Search Terminal Help
Information: Running common shell Library Compiler. (LCSH-1)
Error: Can't open file '/home/b10307123/Desktop/master_v4/modules/SRAM_SP_ADV/SRAM_512x64x8/SRAM_512x64x8_ss_0.9_125.0_syn.lib'. (FILE-1)
Warning: The 'USERLIB' library has not been read in yet. (UIL-3)
0
design_vision> source mem_lib2db.tcl
Information : write_lib mode is turned on already...
Information: Running common shell Library Compiler. (LCSH-1)
Error: Can't open file '/home/b10307123/Desktop/master_v4/modules/SRAM_SP_ADV/SRAM_SP_ADV/SRAM_SP_ADV_ss_0.9_125.0_syn.lib'. (FILE-1)
Warning: The 'USERLIB' library has not been read in yet. (UIL-3)
0
design_vision> source mem_lib2db.tcl
Information : write_lib mode is turned on already...
Information: Running common shell Library Compiler. (LCSH-1)
Reading '/home/b10307123/Desktop/master_v4/modules/SRAM_SP_ADV/SRAM_SP_ADV_ss_0.9_125.0_syn.lib' ...
Technology library 'USERLIB' read successfully
Wrote the 'USERLIB' library to '/home/b10307123/Desktop/master_v4/modules/SRAM_SP_ADV/SRAM_SP_ADV_ss_0.9_125.0_syn.db' successfully.
1
design_vision>
Suspended
[b10307123@EE310 SRAM_SP_ADV]$
```

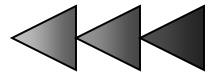
Memory & IC Compiler



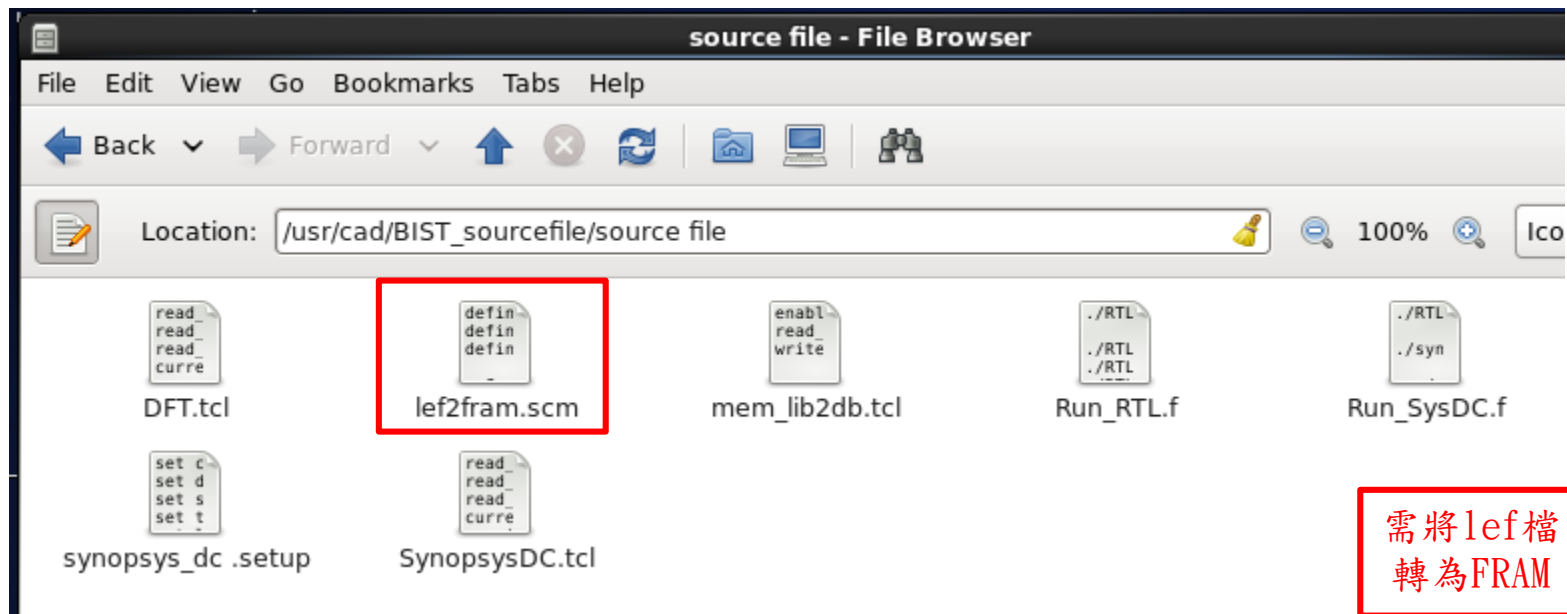
- ❑ In ur memory folder, create another folder called MEM_ICC



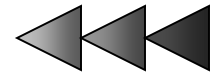
Generate Memory Library



- ❑ Go to `/usr/cad/BIST_sourcefile/source file` and copy `lef2fram.scm` file into ur `MEM_ICC` folder



Generate Memory Library



 Edit the copied file

lef2fram.scm (~/Desktop/layout/SRAM_SP_ADV/MEM_ICC) - gedit

File Edit View Search Tools Documents Help

Open Save Undo

```
define lib_name "SRAM SP ADV"
define tech_file "/usr/cad/CBDK TSMC90G Arm v1.1/icc lib1.1/tsmc090_9lm_2thick.tf"
define data_path "/home/aaazx610/Desktop/layout/SRAM_SP_ADV" |
```

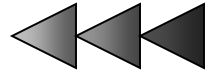
將絕對路徑改為你
vclef檔的絕對路徑

lib_name: The name of the library that ur going to generate

tech_file: the directory of the technology file (.tf)

data_path: the directory of the .vclef file, everyone may be different, so **make sure u set this right**

Generate Memory Library



- ❑ Enter **Milkyway** and wait for a little while:

注意開terminal位置

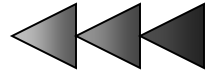
```
aaazx610@EE310:~/Desktop/layout/SRAM_SP_ADV/MEM_ICC
File Edit View Search Terminal Help
[aaazx610@EE310 MEM_ICC]$ tcsh
Platform = amd64
Platform = amd64
Platform = amd64
Platform = amd64
Platform = Linux x86_64
[aaazx610@EE310 MEM_ICC]$ Milkyway
Connection to IPC server failed.
Starting IPC Server. (Port = 1925)
IPC Server started successfully.

Starting a non page replacement mode Milkyway db session...
█
```

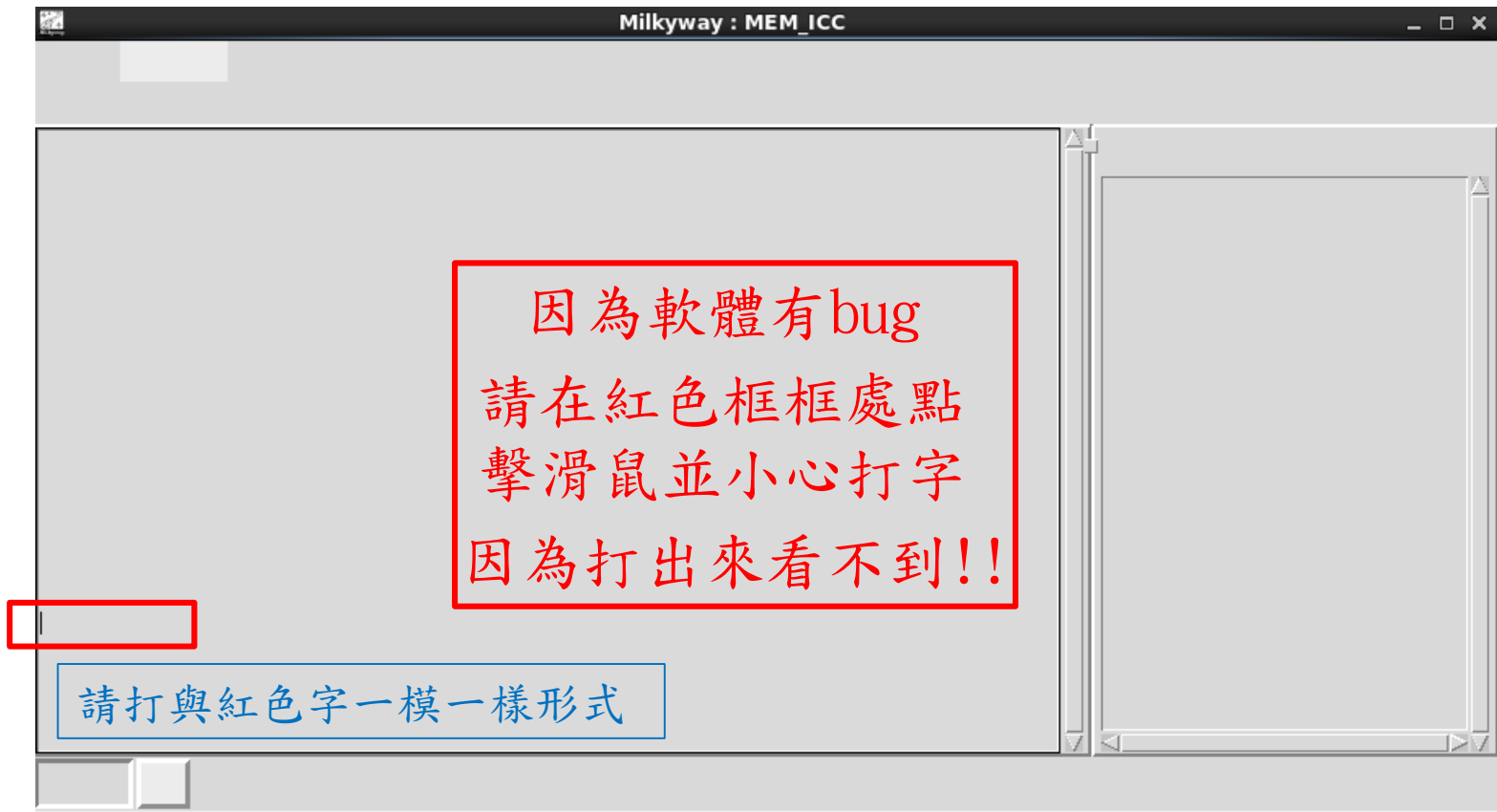
Milkyway -load lef2fram.scm

打這樣可以跳過下一頁

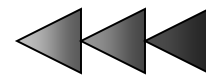
Generate Memory Library



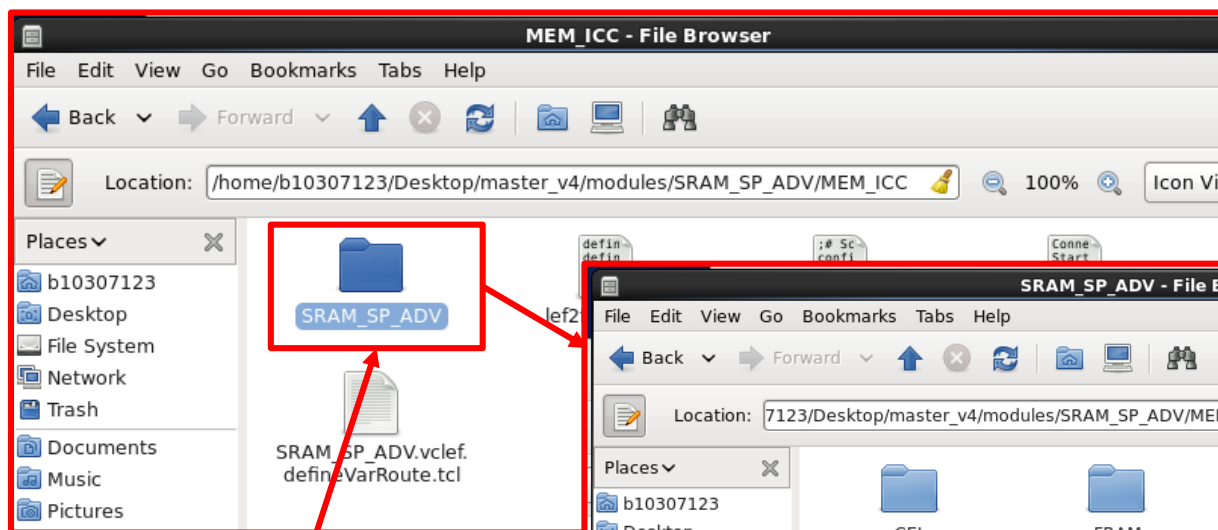
- ❑ Enter **load “lef2fram.scm”**, the entered letters may be **invisible**, so be sure u enter the correct letters:



Generate Memory Library

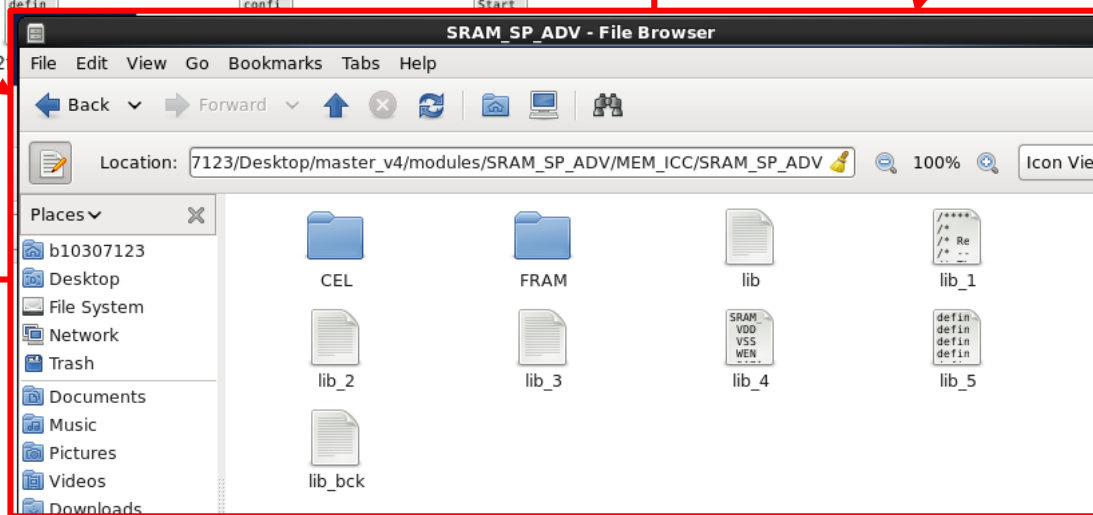


❑ Go into the generated folder to make sure the library is

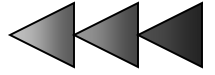


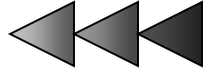
最後產出檔案需如附圖
只要有少就要重產
回去看路徑指令有無問題

在ICC階段會吃
macro的資料夾
就是這個資料夾



Question?





Memory Generator End!